

Dynamic path planning based on the fusion of improved RRT and DWA algorithms

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Abstract—A hybrid algorithm combining improved RRT algorithm and dynamic window method is proposed to solve the path planning problem of robots in the environment of dynamic or unknown obstacles. Firstly, aiming at the problems of many redundant nodes and low efficiency in the RRT algorithm, the target bias sampling and parent node reselection strategies are used to improve them. Secondly, for the problems of poor environmental adaptability and difficulty in taking into account speed and safety in the dynamic window method, a DWA algorithm with fuzzy control of parameters is proposed, which dynamically adjusts the weight of the evaluation function according to the distance of the robot from the target and obstacle. Finally, the above improved RRT algorithm and fuzzy DWA algorithm are mixed to realize dynamic path programming. By comparing the simulation results, the effectiveness of the hybrid algorithm in the complex dynamic environment is verified.

Keywords—Planning; RRT; DWA; Hybrid algorithms

I. INTRODUCTION

The path planning technology of mobile robots is the core of robots to complete autonomous navigation [1]. Path planning is to plan a safe path from the starting point to the target point in a spatial environment. According to the degree of knowledge of spatial environment information, path planning algorithms are divided into global path planning algorithms and local path planning algorithms.

The study of a path planning method that takes into account global path optimization and local random obstacle avoidance has become a research hotspot [2]. Literature 3 proposes an algorithm combining BIT* and DWA, optimizes the path obtained by BIT* algorithm, and improves the evaluation function of DWA [3], but does not optimize and improve the weight of the evaluation function, and the environmental adaptability is insufficient. Literature 4 improves the search point selection strategy and evaluation function of the A* algorithm, and after fusing the DWA algorithm, random obstacle avoidance can be realized [4]. Literature 5 improves the A* algorithm so that the path of a single plan can pass through multiple target points, and at the same time combine with the dynamic window method to achieve the pursuit of dynamic targets [5]. In order to achieve smooth motion and real-time dynamic obstacle avoidance when the mobile robot moves along

the global path, this paper proposes a dynamic path planning algorithm that improves the fusion of RRT algorithm and dynamic window method.

II. IMPROVED RRT ALGORITHM

A. Improve the RRT algorithm process

In view of the problems existing in traditional RRT, this paper proposes two improvements to the RRT algorithm: first, in order to avoid the blindness of sampling and improve the search efficiency of the RRT algorithm, the target bias sampling method is adopted; Second, in order to reduce path redundant nodes and unnecessary turnover, and optimize the path length, a parent node reselection strategy is proposed to optimize the path length.

B. Target bias sampling method

The target bias sampling method is to make the random tree expand to the target point with a certain probability. The specific content is: first, according to the complexity of the map, set the target bias probability threshold Q ($Q \in (0, 1)$), in order to average the probability, the Gaussian random number Q_{rand} ($Q_{rand} \in (0, 1)$) is used to compare with the above probability threshold Q . When $Q_{rand} < Q$, the sampling point is sampled in the direction of the target point; When $Q_{rand} \geq Q$, the sample points are randomly acquired in free space. Through target bias sampling, the expansion of random trees to target points can be promoted, and the search efficiency of the algorithm can be improved.

C. The parent node reselects the policy

After the RRT algorithm obtains the initial path, the initial path often has unnecessary redundant nodes and turns, which increases the path length. This paper proposes a parent node reselection strategy, which shortens the path length by reselecting the parent node of the node in the path, eliminating the redundant node, retaining only the necessary turning point.

The parent node reselection strategy is shown in Figure 1, the specific steps are as follows: set the path node to $\{P_k | k=1, 2, \dots, n\}$, and iterate through all nodes in the path in turn, taking P_5 as an example, let P_4 be the original parent node of P_5 , and P_5 back-detect the previous path node, at this time, P_1 and P_5 can be connected without collision, but if a collision occurs between

Start and P5, the traceback is terminated, the parent node of P5 is replaced by P1, directly connected to P1, P5, and repeats the above operations. until you have traversed all the nodes in the path.

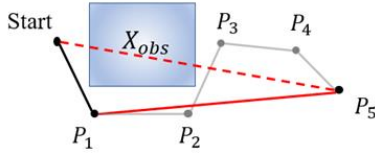


Fig. 1. Parent node reselection policy diagram

After the parent node reselection policy operation, the path planned by the RRT algorithm only includes the starting point, the end point, and the necessary transition node, which effectively reduces the path length

III. ALGORITHM OPTIMIZATION

Dynamic Window Method (DWA) is a common algorithm to solve the local path planning problem of mobile robots. The DWA algorithm determines the range of robot motion speed, infers the feasible motion trajectory of the robot in the next period of time, and then determines the optimal trajectory of the robot through multiple evaluation functions.

A. Classical DWA algorithm

The traditional DWA algorithm comprehensively considers the heading, speed and safety of the robot in the evaluation function, and the evaluation function is defined as follows:

$$G(v, \omega) = \alpha Head(v, \omega) + \beta Vel(v, \omega) + \gamma Dist(v, \omega) \quad (1)$$

The $Head(v, \omega)$ function is the azimuth evaluation function, which indicates the angle between the direction of the end of the trajectory and the target point. $Vel(v, \omega)$ is a speed evaluation function, which represents the current estimated operating speed; $Dist(v, \omega)$ is a distance evaluation function that represents the distance between the trajectory and the nearest obstacle. The evaluation function $G(v, \omega)$ indicates that the robot is moving towards the destination at a large speed while maintaining a safe distance from the obstacle. α , β , and γ are the weighting coefficients of azimuth, velocity, and distance evaluation functions, respectively.

B. Fuzzy DWA algorithm

Traditional DWA algorithms have the disadvantage of fixed weights and cannot adapt to various complex or dynamic environments. In this paper, a fuzzy DWA algorithm is proposed, which dynamically adjusts the weight between each function according to the distance between the robot and the target and obstacle, improves the environmental adaptability of the algorithm, and optimizes the distance evaluation function.

Aiming at the problem that the traditional DWA algorithm uses the same evaluation function for static obstacles and dynamic obstacles in a dynamic environment, this paper subdivides the distance evaluation function, which are $Dist_{SO}$ and dynamic obstacle evaluation function $Dist_{DO}$. If the value of this evaluation function is small, it means that an obstacle is approaching. On the contrary, it means staying away from obstacles.

Let the position of the robot be (x_n, y_n) ; The chassis radius of the mobile robot is R ; The position of static obstacles is (x_{SO}, y_{SO}) ; Let the distance of the robot decelerate at the maximum deceleration at the maximum speed L ; D_{SO} is the minimum distance from the robot to a static obstacle; Then the static obstacle evaluation function $Dist_{SO}$ is shown in Equations 2,3.

$$D_{SO} = \sqrt{(x_n - x_{SO})^2 + (y_n - y_{SO})^2} \quad (2)$$

$$Dist_{SO} = \begin{cases} D_{SO}, & D_{SO} < 1.5L \\ 1.5L, & D_{SO} \geq 1.5L \end{cases} \quad (3)$$

Order to ensure a safe distance between the robot and the dynamic obstacle, the robot is encouraged to stay away from moving obstacles by lowering the score of the trajectory close to the dynamic obstacle. Let the position of the dynamic obstacle be (x_{DO}, y_{DO}) , D_{DO} be the minimum distance from the robot to the dynamic obstacle, and the dynamic obstacle distance evaluation function $Dist_{DO}$ is shown in Equation 4 and 5.

$$D_{DO} = \sqrt{(x_n - x_{DO})^2 + (y_n - y_{DO})^2} \quad (4)$$

$$Dist_{SO} = \begin{cases} D_{DO} - \frac{2R}{D_{DO}}, & D_{DO} < 1.5L \\ 1.5L, & D_{DO} \geq 1.5L \end{cases} \quad (5)$$

The final DWA evaluation function is:

$$G(v, \omega) = \alpha Head(v, \omega) + \beta Vel(v, \omega) + \gamma (Dist_{SO}(v, \omega) + Dist_{DO}(v, \omega)) \quad (6)$$

Equation 6 indicates that the robot is moving towards the target point at a large speed while keeping a distance from static and dynamic obstacles.

Equation 6 indicates that the robot is moving towards the target point at a large speed while keeping a distance from static and dynamic obstacles. The fuzzy language value is $\{O, PS, PM, PB, PH\} = \{\text{"Minimal", "Small", "Medium", "Large", "Maximum"}\}$, and the input and output correspond to the domain and trigonometric membership functions

Let the distance at which the robot decelerate at maximum deceleration be L . When $L_{goal} > L$, $L_{obs} > L$, the robot is far away from the obstacle and does not need to consider avoiding the obstacle. The robot can be adjusted to the appropriate orientation and accelerates closer to the target point. α , β can be selected as a larger value or γ as a smaller value.

When $L_{goal} > L$, $L_{obs} \leq L$, the robot should avoid obstacles before approaching the target point. Therefore, the speed of the robot should be appropriately reduced to reserve time and distance for the robot to avoid obstacles. α , β can select a smaller value γ a larger value.

When $L_{goal} \leq L$, $L_{obs} > L$, the robot can approach the target point first and is not in a hurry to avoid obstacles. Reduce the robot speed appropriately to ensure that the robot does not rush over the target point. α can select a larger value, β can select a medium value, γ select a smaller value.

When $L_{goal} \leq L$, $L_{obs} \leq L$, the robot needs to avoid obstacles first, and then approach the target at a suitable speed and direction. α select a medium value, β select a smaller value, γ select a larger value.

At the same time, in order to obtain more accurate and smooth control variables, the area center of gravity method is used as the fuzzy solution.

IV. ALGORITHMIC FUSION

A. Authors and Affiliations

The RRT algorithm has strong search ability and will not fall into local optimum, but it cannot get a safe path in the face of unknown static obstacles and dynamic obstacles. The DWA algorithm has the advantages of better real-time and in line with the motion characteristics of the robot, but the problems of lengthy routes, falling into local optimum, and unable to reach the target point often occur.

In view of the above problems, this paper combines the advantages of the two algorithms to propose an fusion algorithm based on improved RRT and DWA. Firstly, the improved RRT algorithm is used to plan the global optimal path, and the global path key point is extracted as the intermediate node of the DWA algorithm, and the dynamic window method is used to carry out local path planning between every two adjacent nodes. The basic flowchart of the mixed-path planning algorithm in this article is shown in the figure. The specific implementation steps are as follows.

Step 1: The mobile robot uses the vehicle-mounted sensor to obtain information about surrounding obstacles and build a grid map of the surrounding environment. And initialize the algorithm parameters to determine the position and target point of the robot.

Step 2: In the raster map, determine the start point and target point; Plan a collision-free global path with improved RRT* algorithms.

Step 3: Optimize the nodes in the global path obtained by the improved RRT algorithm to the necessary transition nodes, and record these nodes as local target points.

Step 4: Use the improved dynamic window method to make the robot move between local target points; During this period, the mobile robot needs to use sensors to obtain information about surrounding obstacles and update it in the grid map in real time; If unknown or dynamic obstacles occur, the improved dynamic window method is used to avoid obstacles in real time and move towards local target points.

Step 5: Determine whether the robot has reached the global target point, and if so, jump to step 6; If the global target point is not reached, update the next local target point and adjust it to step 4.

Step 6: The mobile robot reaches the global target point and the algorithm ends.

V. SIMULATION EXPERIMENTS

In order to verify the feasibility and effectiveness of the proposed RRT algorithm combined with dynamic window method in complex environment path planning, the improved algorithm is simulated and verified. Using the open-source tool PathEnv under PYTHON, the simulation environment is a two-dimensional grid map, in which the black grid represents the obstacle area, the white grid represents the movable area, the red

grid represents the movable obstacle, and the gray grid represents the unknown static obstacle.

A. RRT algorithm simulation

Compared with the traditional RRT algorithm, the improved RRT algorithm optimizes the sampling method and optimizes the path by using a parent node reselection strategy. The simulation results are shown in Figure 2, and the performance pair is shown in Tables 1.

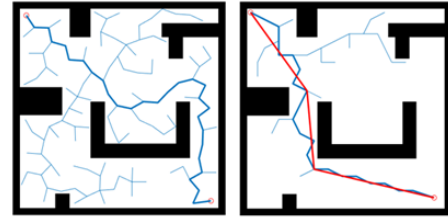


Fig. 2 Map zero Simulation result map

TABLE I. MAP 1 PERFORMANCE COMPARISON

Map 0	Time(s)	Length(m)	Path node
RRT	0.17	52.21	26
Improved RRT	0.12	43.08	4

As can be seen from Table I, compared with the traditional RRT algorithm, the path length of the improved RRT algorithm is shortened by 19%, the planning time is shortened by 26%, and a large number of redundant nodes are removed. The improved RRT algorithm effectively obtains a global navigation path that only contains the starting point, key turning point and target point.

B. Algorithm compared to simulation experiment

In order to verify the effectiveness of the fuzzy DWA algorithm, the traditional DWA algorithm and the fuzzy DWA algorithm are simulated and compared. The weighting coefficients of the traditional DWA algorithm evaluation function are: $\alpha = 0.1$, $\beta = 0.2$, $\gamma = 0.25$.

In order to verify that the fuzzy DWA algorithm can take into account both safety and efficiency, the traditional DWA algorithm and the fuzzy DWA algorithm are simulated and compared.

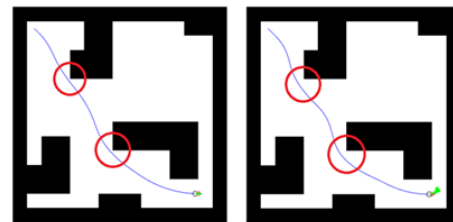


Fig. 3. Map one-weight change plot

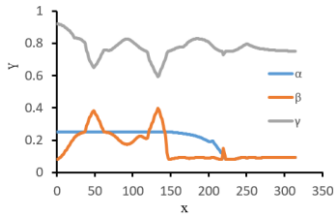


Fig. 4. Map one -weight change plot

TABLE II. TABLE TYPE STYLES

Map 1	Number of iterations	Time(s)	Closest distance to obstacle(m)
Traditional DWA	429	18.32	0.12
Fuzzy DWA	326	12.41	0.26

It can be observed from Figure 3 that the traditional DWA algorithm is only 0.12 meters away from the obstacle when passing through the obstacle, which is less safe; the fuzzy DWA algorithm maintains a long distance from the obstacle while ensuring the speed, and the closest distance to the obstacle is 0.26 meters. As can be seen from Figure 4, the fuzzy rule can dynamically adjust the parameter weights according to the distance between the robot and the target point and obstacle. Table II shows that the number of iterations of the fuzzy DWA algorithm is reduced by 24% and the running time is reduced by 32%.

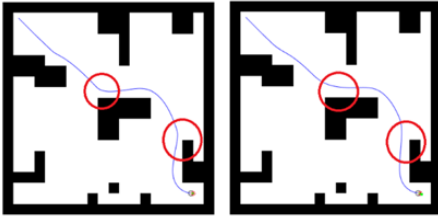


Fig. 5. Map one-weight change plot

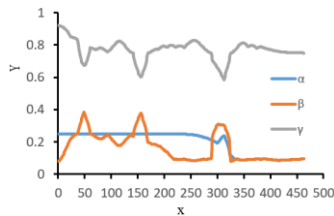


Fig.6. Map one -weight change plot

TABLE III. TABLE TYPE STYLES

Map 1	Number of iterations	Time(s)	Closest distance to obstacle(m)
Traditional DWA	565	29.03	0.52
Fuzzy DWA	478	25.18	0.74

It can be seen from Figures 5 and 6 that the fuzzy DWA algorithm is farther away from obstacles than the traditional DWA algorithm, and the parameter weights can be dynamically adjusted according to the surrounding environment. Table III shows that the iteration number of fuzzy DWA algorithms is reduced by 15.39%, the running time is reduced by 13.26%, and the safety distance is increased by 29.72%. The simulation results fully prove the effectiveness and safety of the fuzzy DWA algorithm in this paper.

C. Fusion algorithm simulation

In order to verify the path planning and real-time obstacle avoidance ability of the fusion algorithm, dynamic obstacles and static unknown obstacles are set on the simulation map. The simulation results of the improved RRT algorithm, DWA algorithm, fuzzy DWA algorithm and fusion algorithm are shown in Figure 7

In the simulation map, the gray grid represents unknown static obstacles, the red grid represents unknown dynamic obstacles, the green dashed line is the path planning path of the improved RRT algorithm, the solid blue line is the traditional DWA algorithm, the solid brown line is the fuzzy DWA algorithm, and the solid red line is the path planning path of the fusion algorithm.

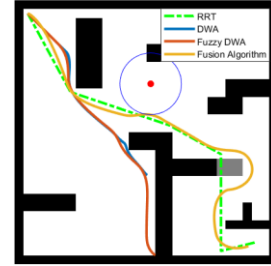


Fig. 7. Map 3 simulation result plot

It can be seen from Figure 7 that under the planning of the fusion algorithm, the path of the robot from the starting point to the target point can effectively avoid dynamic obstacles and unknown static obstacles, and compared with the RRT algorithm, the path of the fusion algorithm is smoother. The experimental results show that by improving the fusion of RRT algorithm and fuzzy DWA algorithm, not only global path planning is realized, but also obstacle avoidance between dynamic obstacles and unknown static obstacles is effectively realized.

VI. CONCLUSION

In this paper, a robot path planning method that integrates the improved RRT algorithm and the fuzzy DWA algorithm is proposed to solve the path planning problem of robots in the environment of dynamic or unknown static obstacles. Compared with the traditional RRT algorithm, the improved RRT algorithm shortens the path length by 19% and the planning time by 26% on average, and removes a large number of redundant nodes, which shows that the improved RRT algorithm in this paper has certain effectiveness and superiority. Compared with the traditional DWA algorithm, the fuzzy DWA algorithm in this paper has significantly improved performance in the number of iterations and running time, and maintains a longer safety distance from obstacles. The fusion algorithm combines the global planning capability of the RRT algorithm with the dynamic planning capability of the DWA algorithm, which ensures the robot's real-time need to avoid unknown moving and static obstacles while having the global path planning capability. After simulation experiments, the path planning and real-time obstacle avoidance ability of the fusion navigation algorithm are verified, which shows that the algorithm is effective and feasible.

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